



K y o t o A I

~Introduction~

CEO Kentaro Seki



0 1 About KyotoAI

0 2 LLM Issues

0 3 Approach

0 4 Research

I n d e x

01

A b o u t K y o t o A I

KyotoAI Inc.

Through fundamental research in AI, we aim to create services that are recognized worldwide and eliminate Japan's digital deficit. Additionally, by developing AI that contributes to research and development in Japan and supporting new industries, we aim to create new employment opportunities through AI.

Name	KyotoAI
CEO	Kentaro Seki
Service	AI Learning, Implementation
Homepage	https://kyotoai.net
GitHub	https://github.com/kyotoai/SEIMEI
Email	office@kyotoai.org
CEO LinkedIn	www.linkedin.com/in/kentaro-seki-b12000339



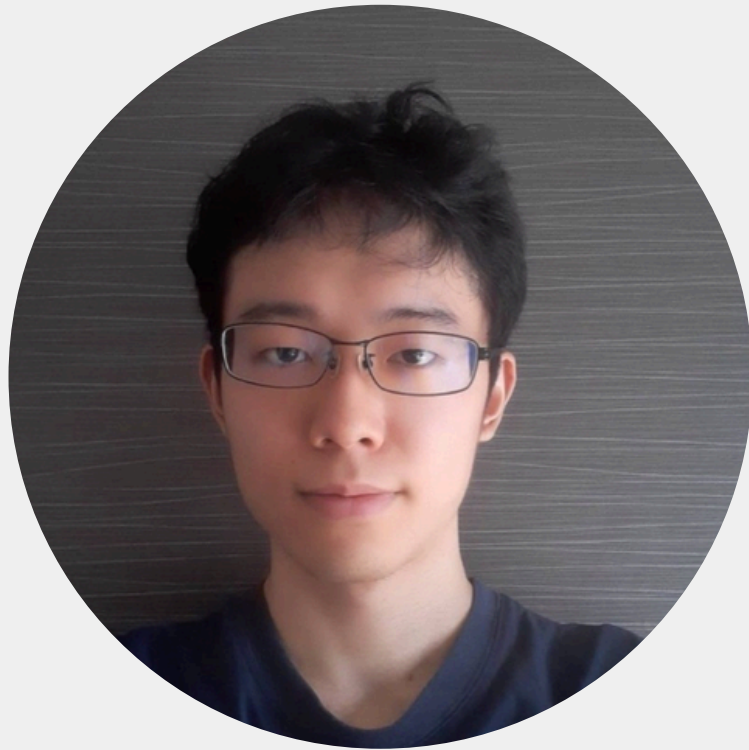
Homepage



GitHub



LinkedIn



○CEO

Kentaro Seki

23 years old

Faculty of Engineering at Kyoto University

AI Engineer, Entrepreneur

Created Chatbot From Scratch

Enrolled in Kyoto University

Taught Myself :

Theory of Relativity, Quantum Physics,
Quantum Field Theory, Particle Physics, Cell
Biology, Neuro Science, Unity, AI Learning,
Philosophy

Created :

Drone from Scratch, Solar Panel, Science
Simulation Game, Monopoly Game, 3D Chess
Game, etc.

KyotoAI :

Started KyotoAI, Fundamental LLM Research,
Proposed a Several New LLM Architectures

M e m b e r s



Mingkwan

○Thai

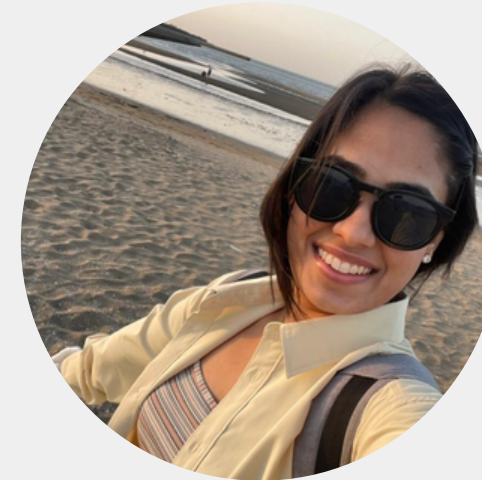
Kyoto Univ
Experienced LLM Intern
Infrastructure/Backend



Kota

○Japanese

IT In a Hotel
Python/Backend



Roshia

○Indian

Kyoto Univ Master
Frontend



Prakhar

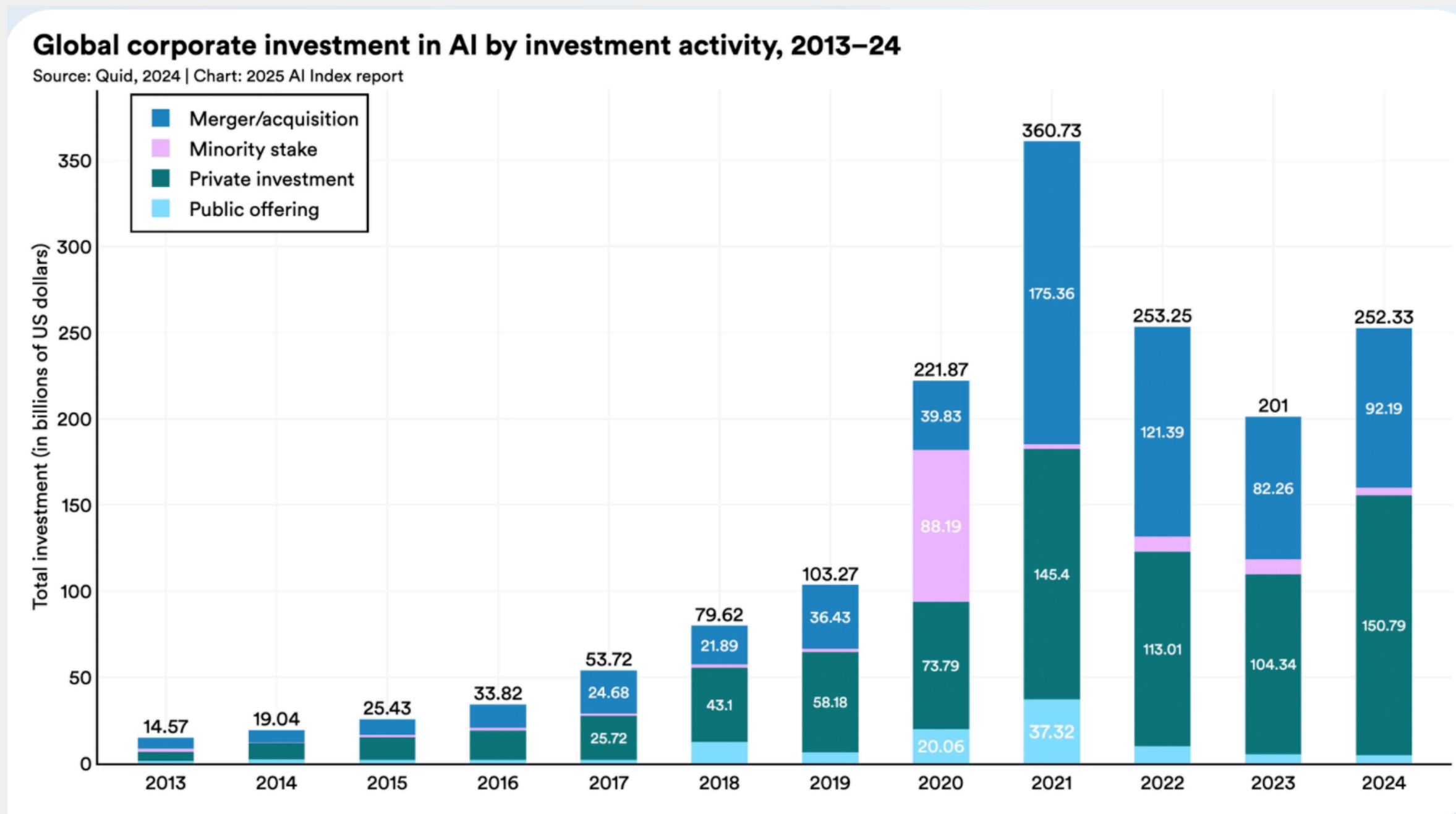
○Indian

Kyoto Univ Master
LLM Researcher
AI Engineer

02

LLM Issues

Unprecedented Investment in LLM



2013~2019:
Ave. \$40~50B



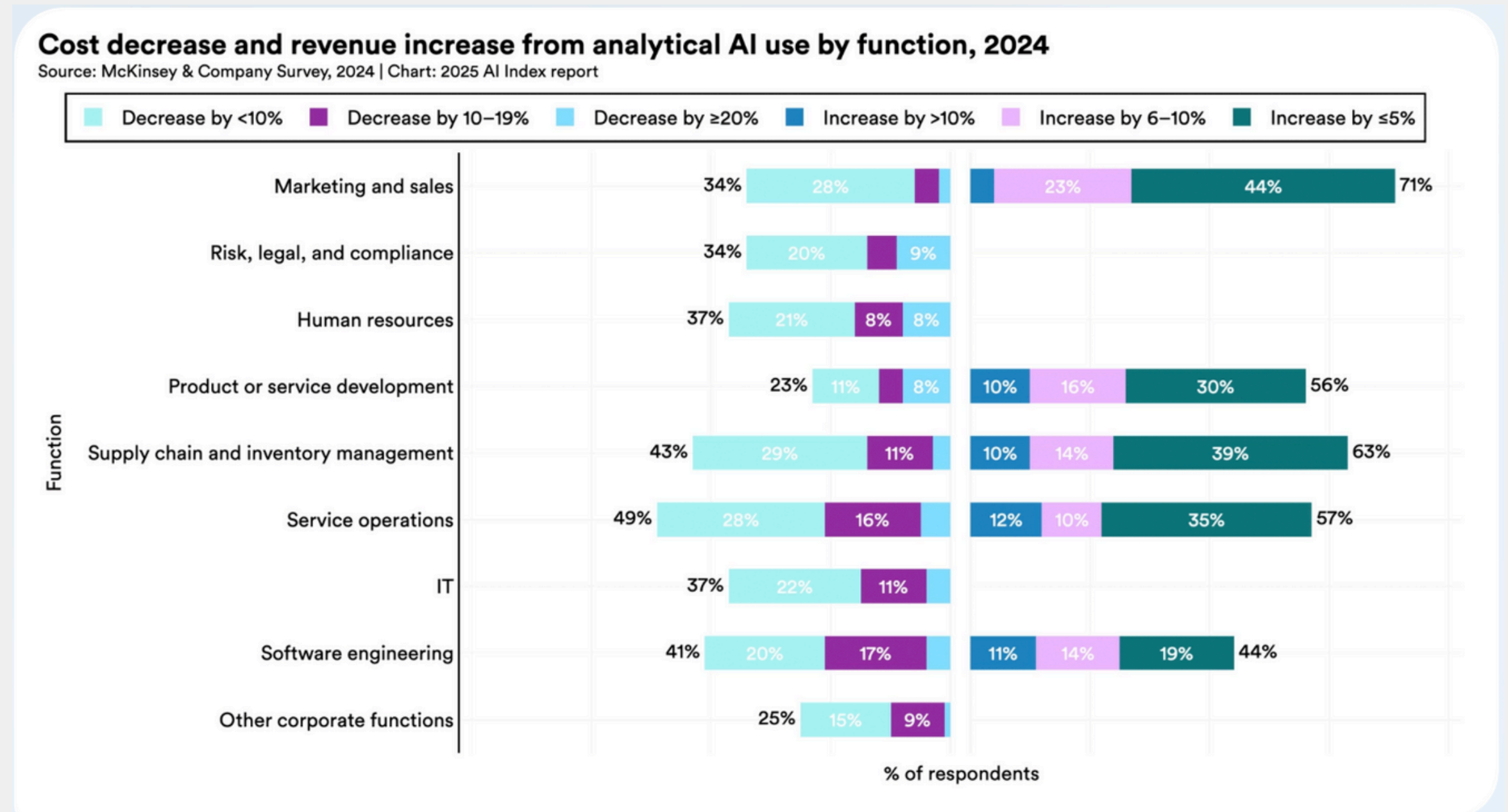
After Generative AI
2020~2024:
Ave. \$250B

Is The Investment Really WORTH ?

Most of the Actual Cases

Cost Decrease < 10%

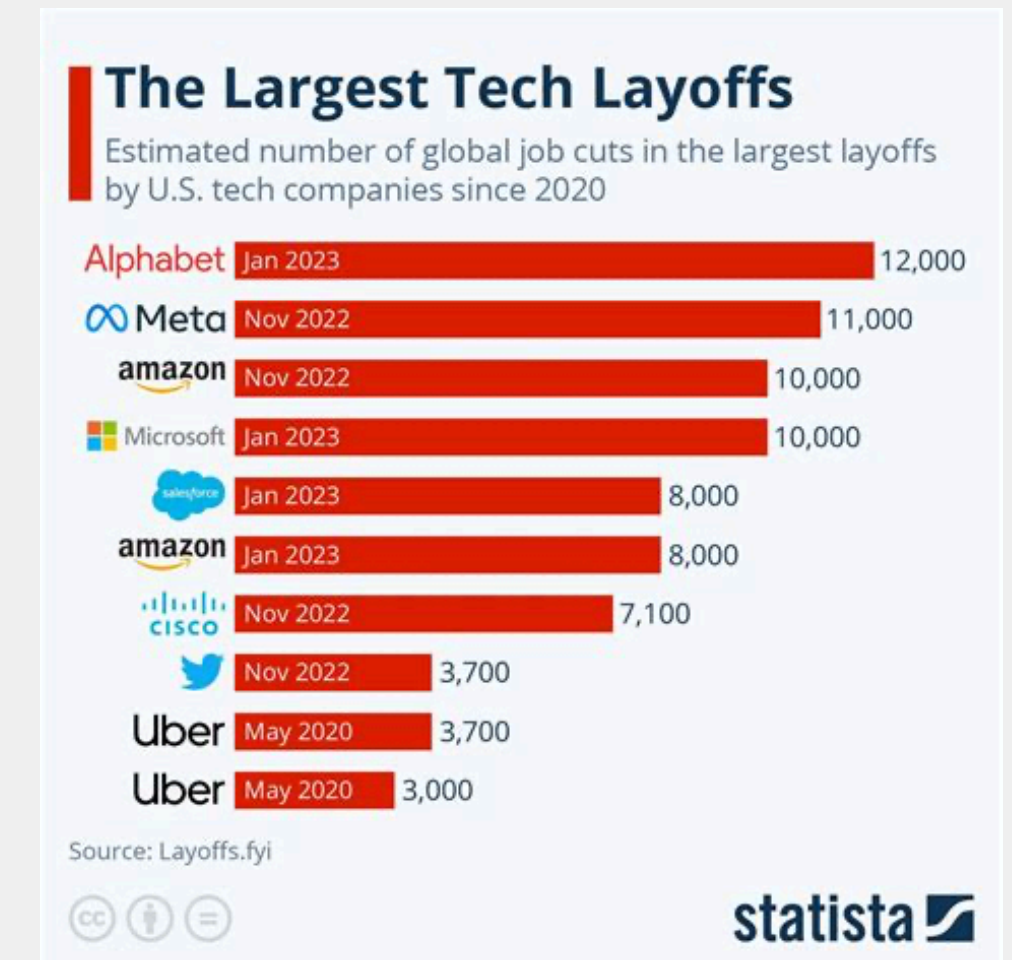
Revenue Increase < 5%



<https://hai.stanford.edu/ai-index/2025-ai-index-report/economy>

Big Layoffs by LLM

- LLM layoffs is becoming a social problem



- Decreasing new employment opportunities

What's wrong with LLM then ?

Large Inovations In The Past Made Impossible Possible

Steam engine ... Can Transport Heavy Things

Airplane ... Enabled humans to fly in the sky

Rocket ... Made expansion into space possible

GPS ... Made it possible to know one's own coordinates

Internet ... Made it possible for people all over the world to connect

On The Other Hand

LLMs are just substitutes for humans for efficiency.

It is not expanding new human possibility.



To Overcome LLM Limitation

AI For Efficiency

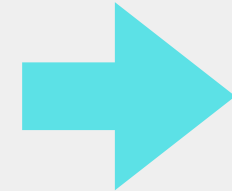
- Tasks that humans can do



Improving work efficiency and digital transformation (DX)



Substitues For Human



AI For Job Creation

- Tasks that humans can't do



Create New Industry



New Innovation Create Jobs

03

A p p r o a c h

KyotoAI Researcher Assistant

```

761
762 -- If we have monitor information then we can offset the mouse by the top-left of the monitor position
763 -- This is because the display-capture source assumes top-left is 0,0 but the mouse uses the total desktop area,
764 -- so a second monitor might start at x:1920, y:0 for example, so when we click at 1920,0 we want it to look like we click
765 if monitor_info then
766     mouse.x = mouse.x - monitor_info.x
767     mouse.y = mouse.y - monitor_info.y
768 end
769
770 -- Now offset the mouse by the crop top-left because if we cropped 100px off of the display clicking at 100,0 should reall
771 mouse.x = mouse.x - zoom.source_crop_filter.x
772 mouse.y = mouse.y - zoom.source_crop_filter.y

```

PROBLEMS 35 OUTPUT DEBUG CONSOLE TERMINAL PORTS

OpenHands

OpenHands CLI v0.47.0

Initialized conversation 2505c5cd-ba74-44f8-9b46-aadb6715694b-25188412acdd5

Let's start building!

What do you want to build? Type /help for help

> コードを解析、ブラウザへのアクセスが可能になるにはどうすればいいか考えよ。ま

まずは
間
魔

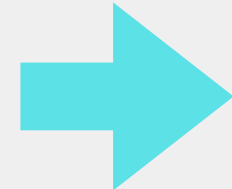
- ✓ Chat with AI via CLI
- ✓ Can be integrated
with VS Code!
- ✓ Custom search model
already integrated

<https://youtu.be/3lQSbQT8KpA>

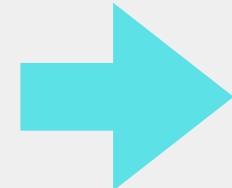
Example Usage



Research Idea



Research Idea



Research Idea



KyotoAI Tool

- Make Branch
- Refer to Papers
- Code Generation
- Debug Code
- Result
- Make README
- Commit Graph
- New Proposal



Next Idea



Next Idea



Next Idea

1. LLM Nuclear Fusion Simulation

Past Fusion Research

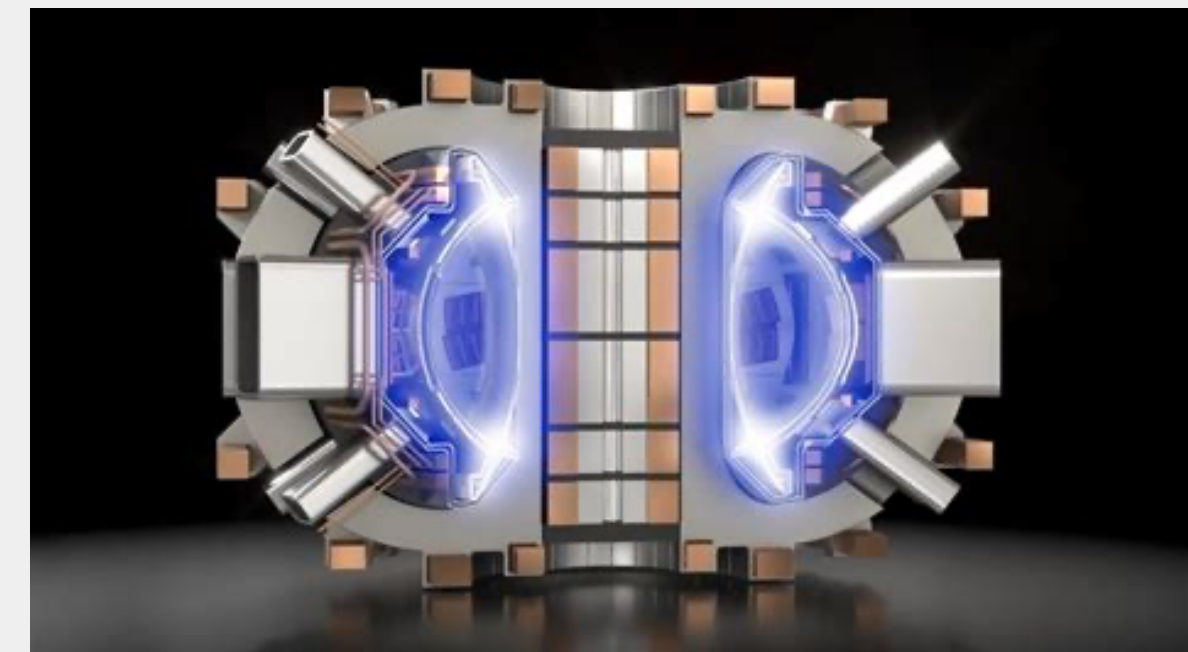
With Nuclear Engineering at Kyoto Univ

- Fusion experts had to write code tailored to their research (taking 1–10 years)
- Countless research topics left unattended due to limited capacity



KyotoAI's Approach

- LLMs generate code and instantly produce results
- Researchers can focus on generating ideas



2. Optimize Communication

With NTT Docomo

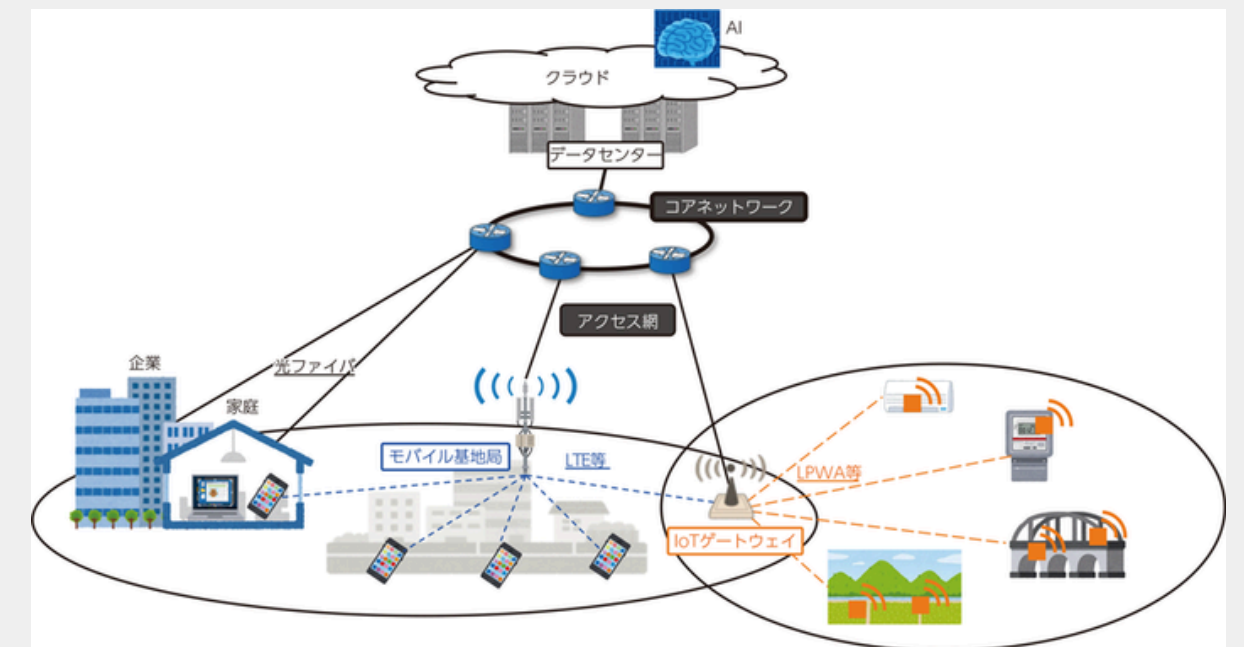
Past Base Station Configuration

- Human experts inferred/decided locations and parameters based on experience
- Areas with poor signal were localized



KyotoAI's Approach

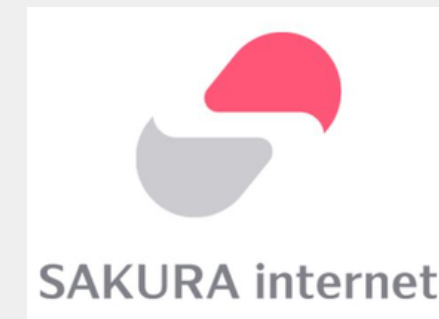
- Analyze and simulate user data with LLMs
- Discover previously unknown relationships



(Idea) Eco LLM with CO2 Free Server

Currently, most LLM servers are located overseas

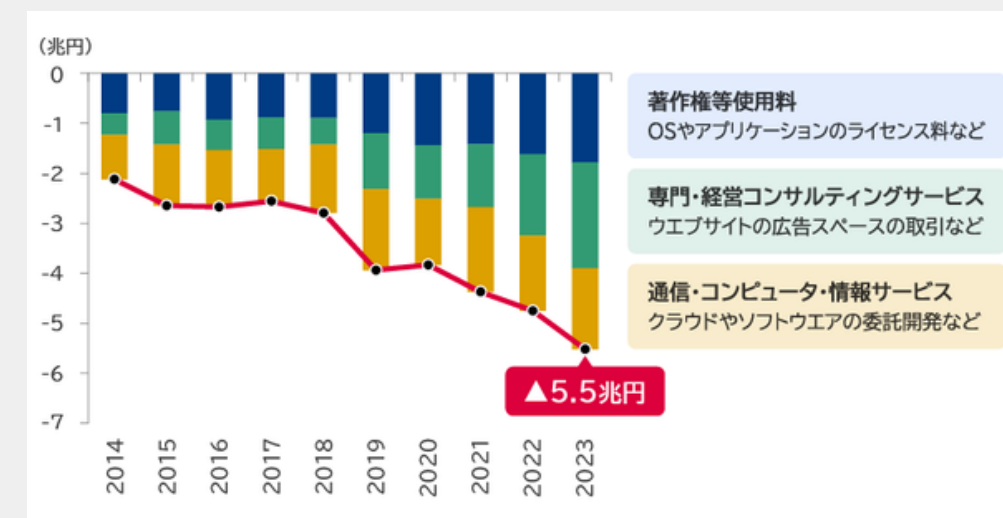
→ Further expansion of the digital trade deficit (5.5 trillion yen)



Domestic & CO2-free — We want to use the Ishikari data center (Sakura Internet)

Domestic & CO2-free & KyotoAI

→ Becomes a major promotional appeal within Japan



Keyword : Inverse Problem

Inverse Problem

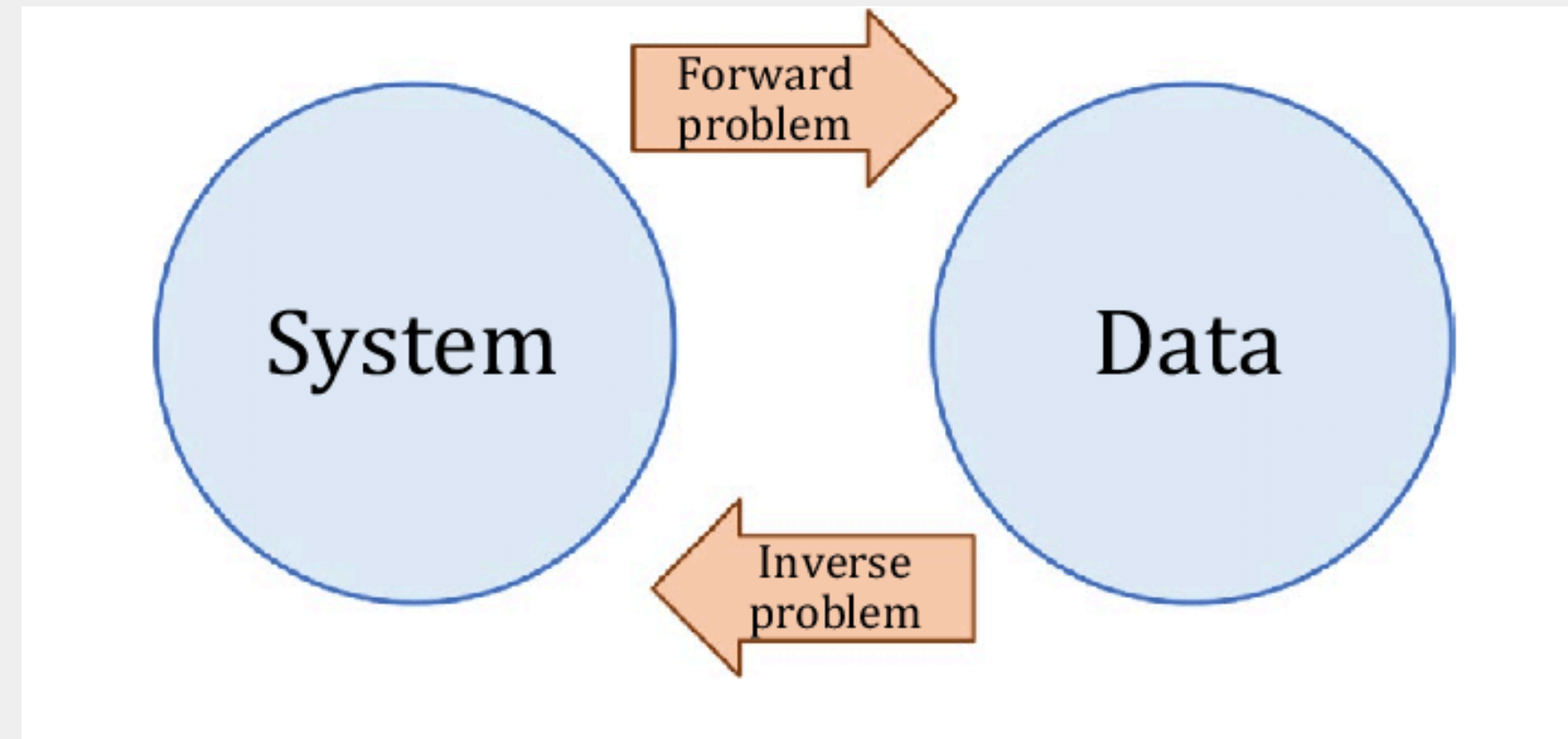
A problem of finding the equation that satisfies the observed data.

In a broader sense,

Visible information → Hidden relationships behind it

is what one seeks to discover—problems that even geniuses can solve only after spending enormous amounts of time.

Ex. new theories in mathematics or physics, creation of new compounds or medicines, and ideas for new inventions.



How to solve Inverse Problems ?

Condition

1. Countless Chains of Thought

Realize nontrivial facts from many thinking experiments

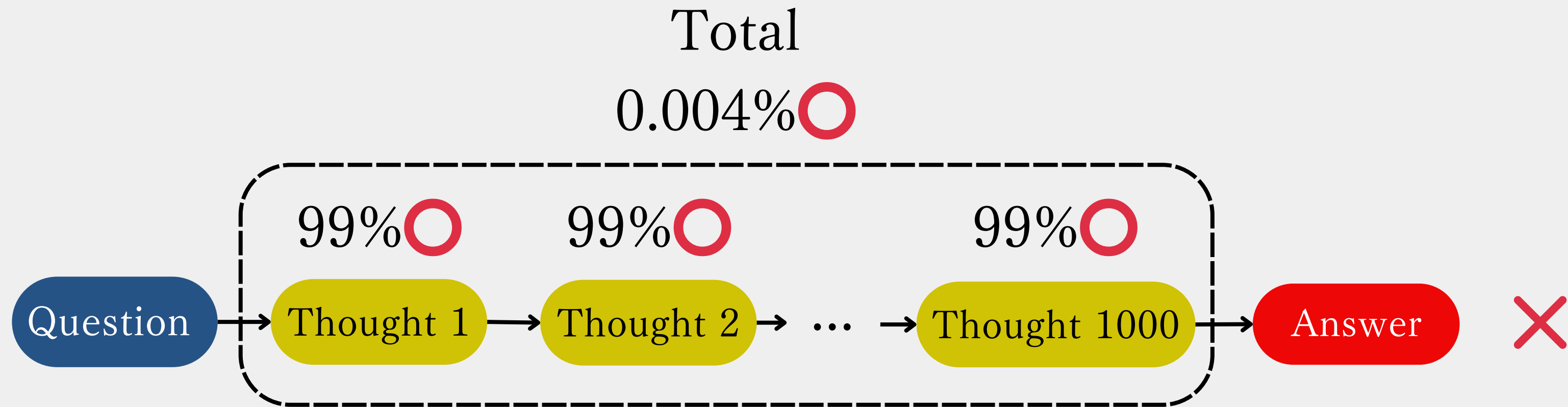
2. Realtime Parameters Update Inside LLM

Improve itself through trials and errors like human

04

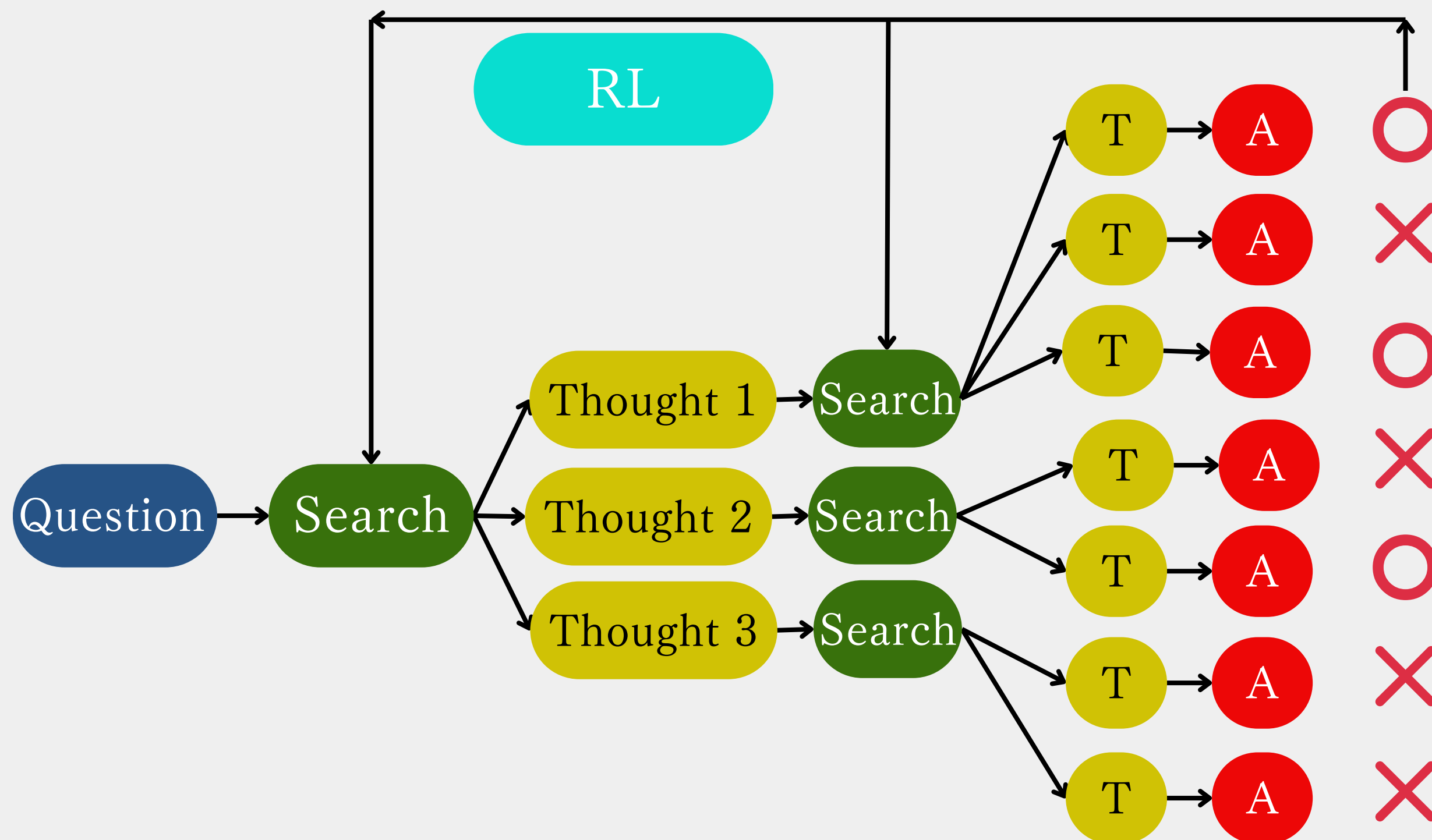
R e s e a r c h

Why can't LLMs think long like humans?



As the thinking continues, it gradually drifts away from what is actually being sought.

Idea : Reinforcement Learning of Search Model

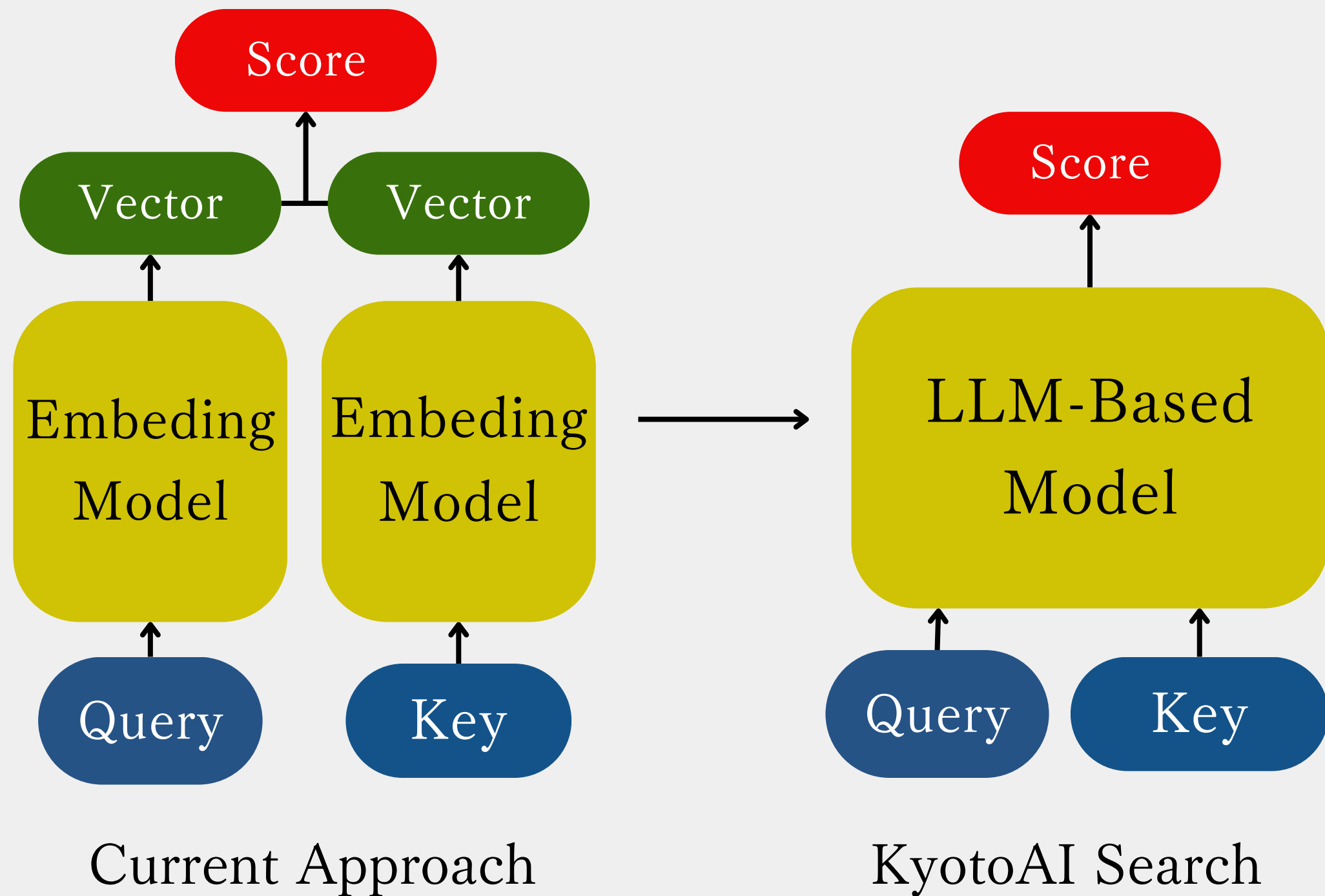


Key Points

1. The search model determines the LLM's thinking steps (similar to human intuition).
2. It guides long-term reasoning toward the thinking steps that lead to the correct answer.

Proprietary Search Model

Remove Vector and Raise Accuracy!



Conventional Model:

- Information content is reduced through vectors
- Requires a database

Original Model:

- Directly calculates scores (higher accuracy)
- No database needed
- Takes more time to compute

Experimental Result of Our Model

Outperforms Current One of the Best Models !

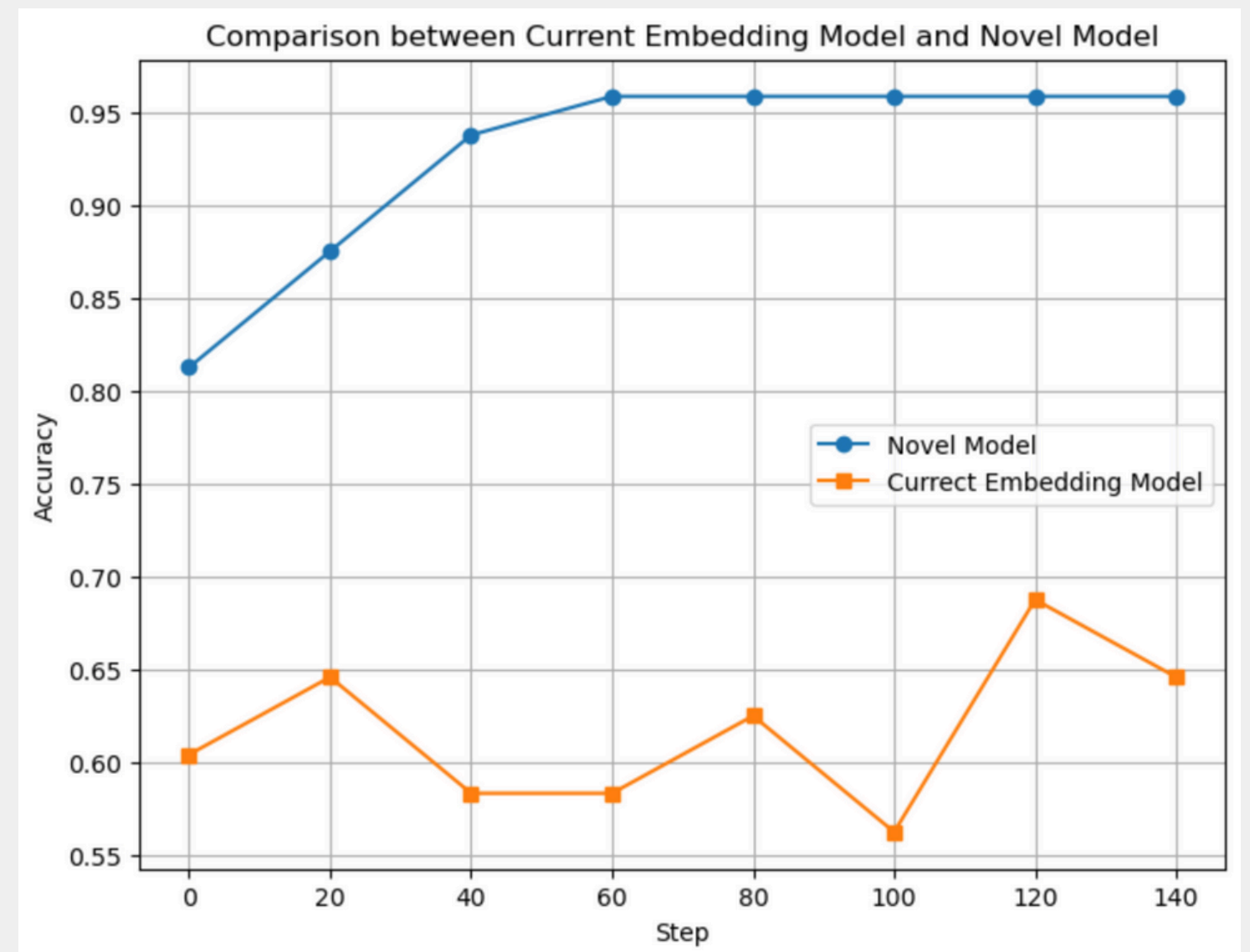
Using the same data, we performed reinforcement learning (DPO) on each model.

DPO Accuracy:

- Conventional Model (■): Does not significantly improve beyond 60%
- Proprietary Model (●): Stable learning, increases up to 96%



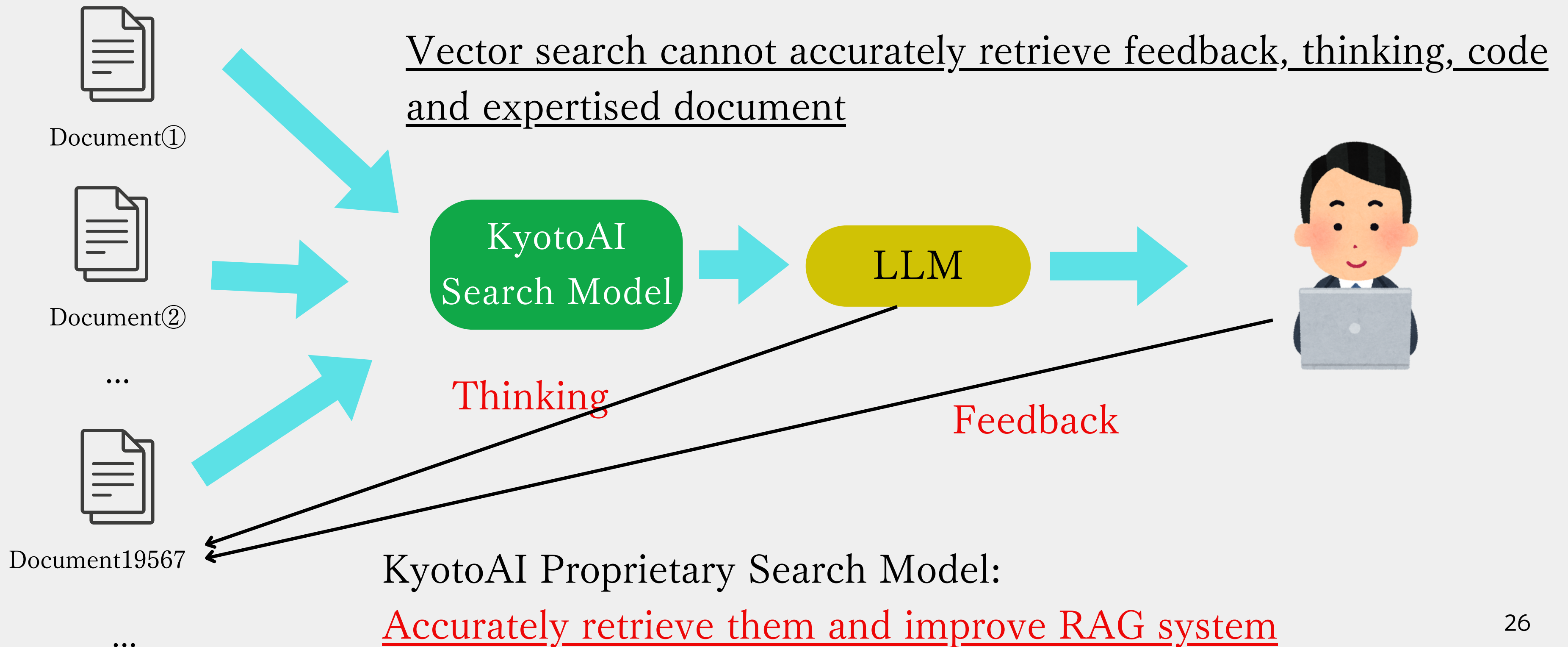
The proprietary model learns more stably compared to the conventional vector search model!



Conventional(■): e5-mistral-7b

KyotoAI(●): 3b-LLM-Based

Ex 1: Realtime Update RAG



Realtime Update RAG: Debug Code Demo

ChatGPT



ChatGPT + Search



ChatGPT: Couldn't remove an error fully

ChatGPT + Search: Could fix the error

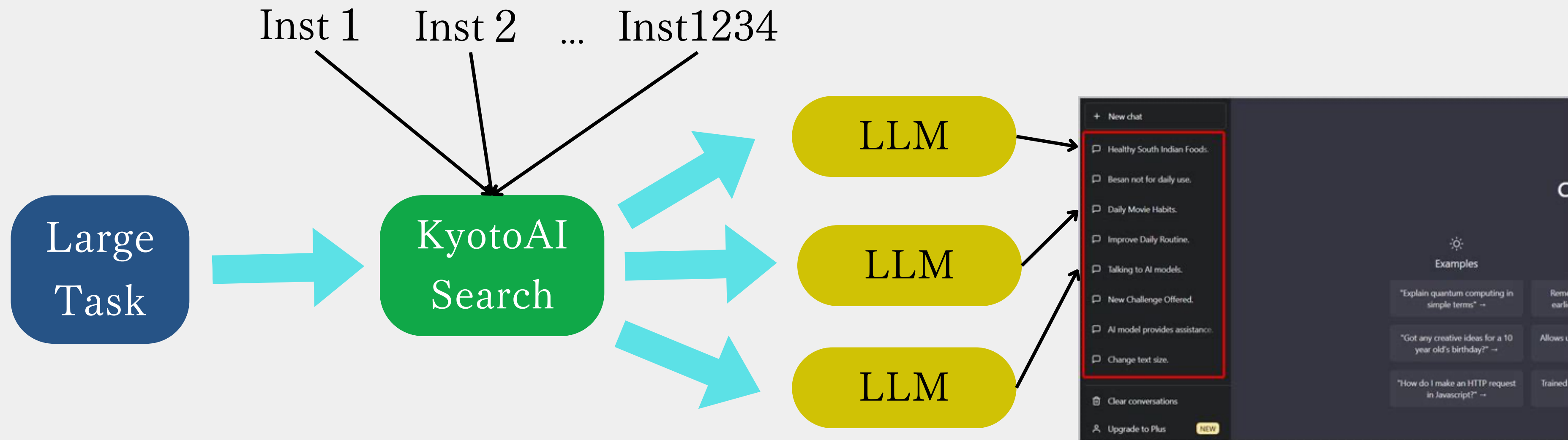
Enhancing ChatGPT with our search model



Now it can solve bug issues that couldn't be removed with ChatGPT alone!!

Reason: The search model looks through code more efficiently
→ pinpoints the location of bugs

Ex 2: Control Instructions → Enable Deeper Thinking



KyotoAI Search Model Controls What To Do Next

→ With the accurate chain of actions, it accurately thinks deeply!!

Result Summary

① Realize High Reasoning like o1 or deepseek
with Search Engine

② x1k~10k Cheaper Training Cost!!

③ Improved accuracy Deepseek-R1/bigcodebench by 16% !

Explore details:

<https://kyotoai-research2503.my.canva.site/en>

